



Equinox Hinge

Self-Locking Composite Deployment

The Equinox Hinge is a next-generation deployment mechanism for CubeSat and SmallSat appendages. Utilizing a specific arrangement of three composite tape springs, it leverages stored strain energy to effect deployment with high torque margins.

Key Advantages: Self-Locking, Shock-Free, High Reliability (No friction/lube), and Zero Backlash.

SPECIFICATIONS

Type	Composite Tape Spring
Material	Woven Carbon Fiber
Mechanism	Strain Energy / Self-Locking
Mass	< 50g
Torque Margin	> 2.0
Pointing	< 0.1°
Op. Temp	-40°C to +85°C
TRL	4-5 (Component)

KEY APPLICATIONS

Solar Arrays	Panel & Wing Deployment
Antennas	Precision Pointing
Instruments	Remote Sensors